P1-18 — Manuscript Readiness Audit and Next Research Priorities

Object: corrected manuscript P1_15_MANUSCRIPT_DRAFT.tex / .md after P1-17, assessed against the frozen blueprint (02_THEOREM_BLUEPRINT.md, SHA-256 ec2fd8d58f8d2a986314f83997d750d689f7c88e98ec8f87753ab4c9cf438cff, unchanged), proof packages P1-04/P1-05/P1-06, the P1-16 audit, the P1-17 correction log, the gap register, and the validation scripts. **Discipline used throughout:** correctness, novelty, significance, and presentation are four different questions. Evidence for one is not evidence for another. No priority or definitive novelty claim is made here.

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1. Repository verification and P1-16 → P1-17 diff

At the start of this phase the repository was synchronized: branch arena/ab8e330d-galios-hull, local HEAD = remote = 1d3242e, working tree clean. The complete commit range 3c72621 (P1-16) → 1d3242e (P1-17) changes exactly six files: the two manuscript files, P1_17_CORRECTION_LOG.md/.pdf, 04_BLUEPRINT_GAP_REGISTER.md, and ARENA_WORKLOG.md. It does **not** touch the frozen blueprint, any proof package, the optional-corollaries file, or any validator.

Every correction claimed in the P1-17 log was verified present in the current text:

- remarks are unnumbered, and a counter simulation gives the P=1 recovery as **Proposition 6.4** in the LaTeX, identical to the Markdown companion; all other numbers are unchanged;
- the Euclidean-dual lemma now contains the r=0 / s=0 boundary verification;
- §2 now defines a λ-constacyclic code over A (shift-closed A-submodule = ideal of $A[x]/(x^n-\lambda)$, no separability/compatibility needed) and Theorem 3.1's closing sentence was corrected accordingly;
- the conditional quantum remark now contains the rank identity $c = \mathrm{rank}(\Pi\,\rho(\Pi)^T) = (n-K) - H$ (kernel argument), cites the Galois form [DIMP2026] within the framework of [BDS2006], and renames the distance d_EA as independently determined;
- [FZ2017] is inserted and cited in §1/§2; the five incomplete bibliography entries are completed; the Talbi citation scope is narrowed to the chain ring; ZKZ2026 initials are added.

No unintended mathematical change was found. No theorem or corollary statement (§§3–9) was altered in substance: the only statement-level touches are the definitional correction in Theorem 3.1's consequence sentence, a shorthand clarification in Theorem 5.2, boundary cases inside one proof, and the §10 conditional remark. The frozen blueprint's SHA-256 is identical before and after.

Structural checks (this phase, on the corrected .tex): braces balanced; environments matched; 33 unique labels with no missing reference; 16 bibliography entries, all cited, none undefined. **Validators re-run:** validate_p1_05.py, validate_p1_06.py, and validate_p1_14.py all exit 0 (ALL … CHECKS PASSED); the P1-14 output is again byte-identical to its saved record. These finite/symbolic checks support — they do not replace — the proofs.

2. What the paper actually proves

2.1 Standing hypotheses (every hypothesis is load-bearing)

$A \cong \prod_{s=1}^{N} K_s$, $K_s = F_{q^{m_s}}$ (square-free affine algebra, fixed labeled components); common $n = P \cdot n_0$ with $P = p^{\varrho}$ and $\gcd(n_0,p)=1$; the code-first k-Galois pairing $\langle x,y\rangle_k = \Sigma\, x_i\, \sigma(y_i)$ with $\sigma = p^k$, $\rho = \sigma^{-1}$; **compatible twists** $\lambda_s^{1+p^{d_s-k}} = 1$ wherever the dual/hull are placed in the same modulus; counting over **labeled** exponent assignments $a_f \in \{0,\dots,P\}$.

2.2 The proved chain

1. **Parametrization (T1–T2).** Component codes are exactly $\langle \prod f^{a_f} \rangle$ for the distinct irreducible factors f of the separable $x^{n_0}-\mu_s$; the Frobenius-twisted reciprocal # is

multiplicative and induces the involution τ_s on factors under compatibility.

2. **Dual and hull in exponent form (T3–T4).** $D(C(a)) = \langle h_a^{\#} \rangle$ with dual exponents $\beta_i = P - \alpha_{i-1}$ (predecessor orientation fixed by the code-first pairing), and $\mathrm{Hull} = \langle \mathrm{lcm}(g_a, h_a^{\#}) \rangle$, giving the local formulas $\dim C = \Sigma\, w_O\, \Sigma_i\, (P-\alpha_i)$, $\dim \mathrm{Hull}_k = \Sigma\, w_O\, \Sigma_i \min(P-\alpha_i, \alpha_{i-1})$ over F_q.

3. **Transfer-matrix enumerator (T5–T6, assembly).** Each τ_s-orbit contributes $\mathrm{tr}(T^{(P)}_{w_O}(u,z)^{L_O})$ with $T_{a,b} = u^{w(P-b)}\, z^{w\cdot\min(P-b,a)}$, and the global joint enumerator is the product over components and orbits: $E(u,z) = \Sigma_{\text{labeled a}}\, u^{\dim C(a)}\, z^{\dim \mathrm{Hull}_k(a)}$. At $P=1$ this is the binary simple-root specialization (Proposition 6.4).

4. **Counting corollaries (C1–C4).** Total $(P+1)^{\Sigma|F_s|}$; code-dimension marginal $\prod_f (1+u^{w_f}+\ldots+u^{Pw_f})$; **LCD count $2^{\#\text{orbits}}$, independent of P** (each orbit contributes exactly its two constant tuples); self-orthogonal $\iff \alpha_{i-1}+\alpha_i \geq P$ on every edge and dual-containing $\iff \leq P$, counted by closed walks of the associated 0–1 matrices, the two counts equal.

5. **Moments (C5–C6).** Under the uniform labeled model: $E[\dim C] = (n/2)\Sigma_s m_s$; $E[\dim \mathrm{Hull}] = \nu_P\cdot(\Sigma_{L=1}\, w_O) + \mu_P\cdot(\Sigma_{L\geq 2}\, w_O\, L_O)$ with $\mu_P = P(2P+1)/(6(P+1))$ and the genuinely different length-one constant $\nu_P = \lfloor P^2/4 \rfloor/(P+1)$; $\mathrm{Var}(\dim \mathrm{Hull}) = \Sigma\, w_O^2\, V_{L_O}(P)$ with three distinct regimes ($L=1$ parity-split, $L=2$, $L\geq 3$ via the negative adjacent covariance κ_P).

2.3 What remains conditional or outside the theorem

- **Incompatible twists:** only the two-modulus dual statement survives; no hull enumerator is asserted.
- **EAQECC remark:** conditional and finite-field only; the rank count is proved inline, but the quantum code exists only via the cited Galois-form construction and an independently determined distance d_{EA}. The enumerator alone yields no distance and no quantum parameter.
- **Moments:** statements about the uniform distribution on labeled assignments, not about codes up to equivalence/isometry.
- **N1–N3 (candidate corollaries, kept separate by instruction):** $\mathrm{Cov}(\dim C, \dim \mathrm{Hull})=0$ (uncorrelated, not independent), $\mathrm{Var}(\dim C)= [P(P+2)/12]\cdot\Sigma\, w_f^2$, and the LCD generating refinement $\Sigma_{a:\mathrm{Hull}=0}\, u^{\dim C(a)} = \prod_{s,O}(1+u^{P\, w_O\, L_O})$ (of which the manuscript's C3 is the $u=1$ count). These are proved and checked in P1_15_OPTIONAL_COROLLARIES.md / P1-14, but they are **not** results of the paper until the authors adopt them; the main text's "LCD refinement" is currently only the count, not the generating function.

3. Readiness evaluation (four separate axes)

Axis | Verdict | Basis

Mathematical correctness | Strong | Complete proof packages for every statement; two independent audits (P1-14, P1-16) with hand re-derivation of the delicate points (predecessor orientation, L=1/L=2 regimes, covariance cancellation, w_O^2 weights); exact validators re-run green this phase; formulas fenced by the correct hypotheses. Residual risk is presentational (compiled-output check), not mathematical, as far as internal review can tell.

Literature novelty | **Unresolved** | The closest item, L1 (Debnath–Prakash–Islam 2023), is verified only at abstract level ("Galois hull dimension … and the number of such codes with a given Galois hull dimension … considering some restrictions on q"); its theorem scope — hypotheses, `p | n` treatment, counting scope, any moment or joint statement — is unknown because the full text has never been lawfully available in this project. Within the ∗inspected∗ sources (affine-algebra hulls DIMP2026, multi-twisted GPM ES2026, average dimensions DP2025, small dimensions DIYP2026, chain-ring hulls Talbi2021, and the Check-D neighbours), no counterpart of the joint affine repeated-root enumerator with moments was found. "No counterpart found in inspected text" is **not** a novelty proof and must not be upgraded into one.

Significance | Moderate, specialized — defensible if positioned honestly | The contribution is a complete exact joint distribution (not just a hull formula for one code) in the repeated-root affine setting, with an orbit-local transfer mechanism, a clean LCD count independent of P, and exact first/second moments including the subtle ν_P correction. Against that: the method (transfer matrices/closed walks) is a standard enumerative technique once the local formula is known; there are no distance results, no new code constructions, and the examples are small. This is a coherent specialist-journal paper, not a breakthrough claim; its strength is completeness and exactness in a regime (repeated root × product alphabet × Galois hull) that the verified neighbours treat only partially.

Presentation | Good draft, not yet a submission | Logical arc (conventions → dual → hull → enumerator → corollaries → moments → examples → limitations) is clear; scope is honestly fenced; examples E1–E5 include field-level certificates. Blockers: author placeholder; never compiled (see §5); the Introduction's closest-neighbour discussion is necessarily abstract-level for L1; N1–N3 undecided; no target-journal formatting.

Overall: the manuscript is ready for author-side compilation and reading, and its mathematical core is in good shape. It is **not submission-ready**: the unresolved L1 overlap is the single largest scientific risk (it bears on novelty and on how the Introduction may position the work), followed by compilation.

4. Bibliography: what each entry is asked to support, and on what evidence

"Inspected" below means: P1-17 primary-record inspection (publisher/Crossref/arXiv/MDPI/Science pages, mostly metadata+abstract) and/or the repository's earlier full-text work (P1-01/P1-09). Nothing is claimed beyond the level stated.

Entry | Claim it supports in the manuscript | Evidence actually inspected | Status

BDS2006 | EA framework reference in the conditional remark | Science journal page + arXiv record (P1-17) | Supports the EA framework; does **not** by itself support the Galois parameter form (that rests on the inline rank proof + DIMP2026).

CDL2015 | Member of the "substantial literature" on repeated-root codes/duals | Crossref record + ScienceDirect page (P1-17) | Bibliography verified; only a literature-existence claim is made.

DP2025 | "Average Galois hull dimensions … computed" (Introduction) | Publisher/abstract level (repository record) | Supports the sentence at abstract level; also why the hull ∗mean∗ cannot be the paper's headline.

DIMP2026 | Affine-algebra Galois hull formulas; simple-root emphasis with repeated-root case under extra hypotheses; Galois-form EA usage in §10 | arXiv full text (P1-01/P1-09); publisher metadata (repo) | Supports these claims as arXiv statements; publisher theorem numbering never checked and never cited.

DIYP2026 | "Constacyclic codes with small Galois hull dimension were studied" | Publisher page/abstract level (repo) | Supports at scope level only.

DPI2023 (L1) | "Determined the Galois hull dimension … and the number of such codes with a given Galois hull dimension" | Abstract/metadata only (Springer/ACM/OpenAlex; repeatedly re-checked in-repo) | **Theorem-level overlap unresolved.** The sentence stays within the abstract; no non-overlap is claimed or claimable.

DPI2023corr | "(see also the correction)" | Correction read in repo work | Supports existence/scope of the correction only.

ES2026 | Generator polynomial matrices for Galois hulls of multi-twisted codes; constacyclic = index one | arXiv full text (P1-01); ScienceDirect metadata + author form (P1-17) | Supports the claim; numbering cited only in arXiv form where used internally.

FZ2017 | Origin of the k-Galois inner product (Introduction, §2) | Springer article page + citing-paper abstracts (P1-17) | Supports the attribution of the pairing $\Sigma\, x_i\, y_i^{p^k}$; the manuscript's code-first dual convention remains its own, not attributed to FZ.

HP2003, MS1977 | "Standard terminology … follows" | Standard textbooks; not re-opened this project | Low-risk background citation; no theorem is rested on them.

Sharma2015 | Repeated-root literature (lengths $\ell^t p^s$, duals) | Springer article page (P1-17) | Bibliography verified; literature-existence claim only.

Talbi2021 | Galois hulls of cyclic serial codes over a finite chain ring | arXiv record (P1-17) | Supports the narrowed Introduction sentence.

WZLL2023 | Repeated-root literature (length $6\ell^m p^n$, duals/LCD/self-dual in scope) | AIMS publisher page incl. abstract (P1-17) | Bibliography verified; literature-existence claim only.

ZKZ2026 | "Galois hulls have likewise been used for quantum-code constructions" | MDPI article page incl. abstract (P1-17) | Supports the existence claim; not used for the manuscript's EA parameters.

ZWLL2025 | Repeated-root literature (length $k^s\ell^m p^n$) | ScienceDirect page (P1-17) + Crossref | Bibliography verified; literature-existence claim only.

Unresolved comparisons: L1 (theorem level) is the only one that can change the paper's positioning; publisher-vs-arXiv numbering for DIMP2026/ES2026 is immaterial because the manuscript cites no theorem numbers from them.

5. Compilation

No LaTeX compiler exists in this environment (pdflatex, xelatex, latexmk, tectonic all absent, re-checked this phase); the manuscript has therefore **not** been compiled, and this report does not

claim otherwise. On the author's TeX system:

```
pdflatex -interaction=nonstopmode P1_15_MANUSCRIPT_DRAFT.tex
pdflatex -interaction=nonstopmode P1_15_MANUSCRIPT_DRAFT.tex   % second pass for \ref
```

(only geometry, amsmath, amssymb, amsthm are required.)

Likely source-level issues the structural checks cannot detect: overfull lines in the long Introduction sentences and in Example 9.3's aligned display; the optional-title rendering of unnumbered remarks ("Remark (conditional quantum remark)."); 2^{\#\mathrm{orbits}} spacing; math inside hand-written bibliography titles (journal styles may restyle \(2\ell^m p^n\) etc.); the draft date and author placeholder; and confirmation in the compiled output that the P=1 proposition indeed prints as Proposition 6.4 with no stray numbered remarks. Two compilation passes are needed before judging cross-references.

6. Smallest set of remaining actions, in priority order

1. **(Scientific blocker) Close the L1 question.** Lawfully obtain Debnath–Prakash–Islam 2023 in full and answer the standing L1-Q1–Q6 (hypotheses; p|n/repeated-root treatment; counting scope — labeled assignments?; any joint (dim, hull) statement; any means/variances). Then update the overlap matrix and, only if warranted, the Introduction's positioning. This is the most consequential remaining research task; everything about novelty framing waits on it.
2. **(Author) Compile and proofread** with §5's instructions; fill authors/affiliations; verify Proposition 6.4 and references in the compiled PDF.
3. **(Author decision) N1–N3 adoption.** Adopting N3 (LCD generating refinement) and N1+N2 (complete second-order picture) would measurably strengthen §7–§8 at low cost; rejection is also safe since nothing depends on them.
4. **(After 1) Position and pick the venue.** Rewrite the closest-work paragraph using verified theorem-level comparisons only, choose the target journal, and apply its bibliography style.
5. **(Optional, later — not required now)** Research extensions the draft already names (incompatible-twist hull intersection; equivalence-class counting; distance-aware refinements) are new projects, not readiness fixes, and should not delay 1–4.

7. Files and blockers

- Created: P1_18_READINESS_AND_RESEARCH_PRIORITIES.md (this report; PDF rendered from it for reading).
- Updated: 04_BLUEPRINT_GAP_REGISTER.md, ARENA_WORKLOG.md (P1-18 outcome).
- Unchanged: manuscript, frozen blueprint (hash in §1), proof packages, optional corollaries, validators.
- Remaining blockers: L1 full text unavailable (novelty unresolved); no LaTeX compiler in this environment; author placeholder and N1–N3 decision outstanding.